Since the value of *k* comes out to be nearly constant, the given reaction is of the first order. The mean value of $k = 0.0676\ min^{-1}$

29.**Solution:**

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A]}$$

$$k = \left(\frac{2.303}{t}\right) \log \left(\frac{V_0}{V_t}\right)$$

In the present case, $V_o = 46.1$ ml.

The value of *k* at each instant can be calculated as follows:

t (*min*)	V_t	$k = \left(\frac{2.303}{t}\right) \log \left(\frac{V_0}{V_t}\right)$
10	29.8	$k = \frac{2.303}{10} \log \left(\frac{46.1}{29.8}\right) = 0.0436\ min^{-1}$
20	19.3	$k = \frac{2.303}{20} \log \left(\frac{46.1}{19.3}\right) = 0.0435\ min^{-1}$

Thus, the value of *k* comes out to be nearly constant. Hence it is a reaction of the first order.

30. Solution:

i) For the first order reaction $k = \frac{2.303}{t} \log \frac{[A_0]}{[A]}$

Assume, $[A_0] = 100\ \%$, $t = 50$ minutes

Therefore, $[A] = 100 - 40 = 60$

$k = (2.303 / 50) \log (100 / 60)$

$k = 0.010216\ min^{-1}$

Hence the value of the rate constant is $0.010216\ min^{-1}$

ii) t = ?, when the reaction is 80% completed,

$[A] = 100 - 80 = 20\ \%$

From above, $k = 0.010216\ min^{-1}$

$t = (2.303 / 0.010216) \log (100 / 20)$

t = 157.58 min

The time at which the reaction will be 80% complete is 157.58 min.

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